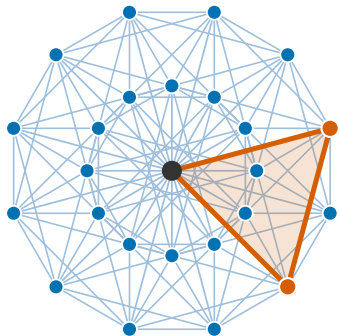
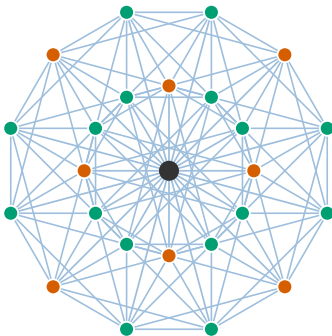


(a) D_4 : 24 links, 96 triangles



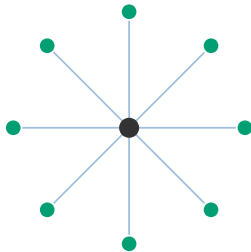
chords = elementary plaquettes
 $\text{Tr} U_{\Delta} = \text{Tr}[U_e U_{f-e} U_{-f}]$

(b) D_4^* : the same 24 links



8 axial + 16 diagonal is a
coordinate split, not a gauge-invariant one

(c) $\mathbb{Z}^4$: 8 links, no triangles



no chords: the lattice is bipartite
the minimal loop needs 4 links